



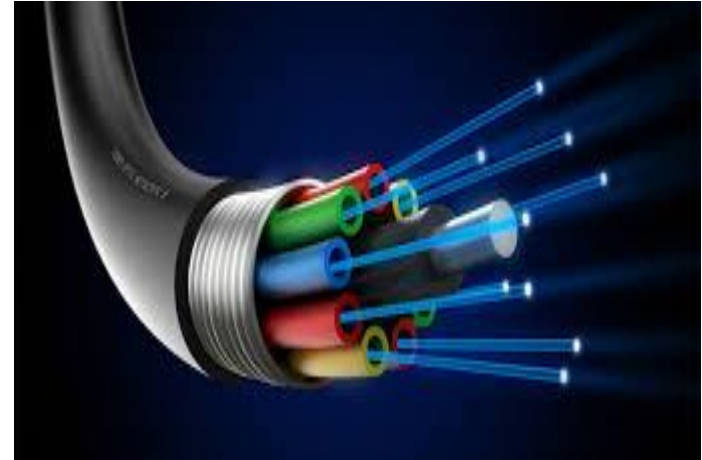
Optical Fiber

Dr. Ashish Bodhaye
Assistant Professor

**St. Vincent Pallotti College of
Engineering & Technology,
Nagpur**

Fiber Optics

Fiber optics is a technology in which electrical signals are converted into optical signals and transmitted through a thin glass fiber and reconverted into electrical signals.

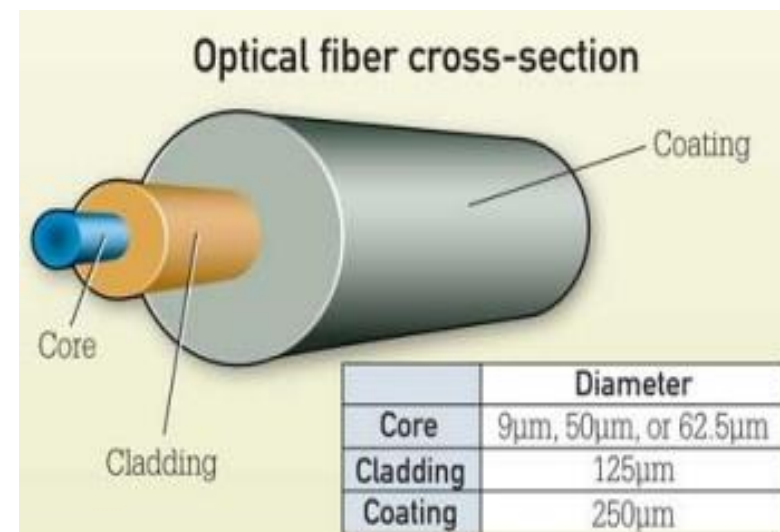


Optical Fiber:

An optical Fiber is a cylindrical wave guide made of transparent glass or plastics, designed to guide light waves along its length by total internal reflection with very little leakage.

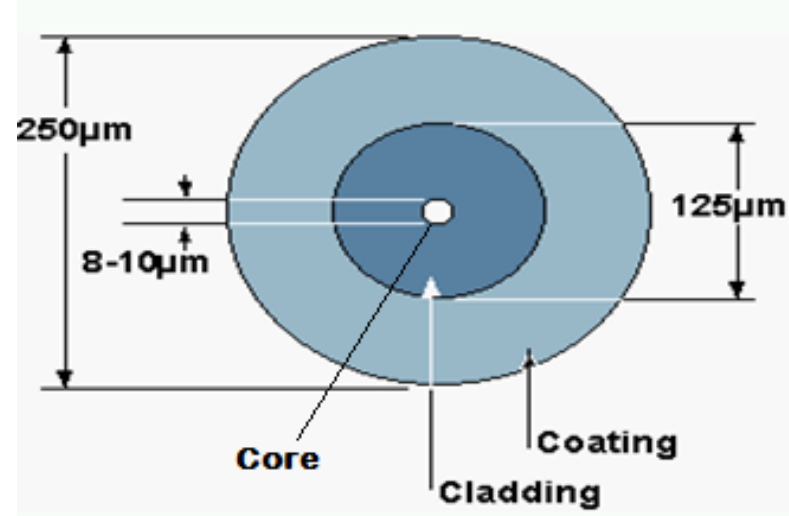
Structure of Optical Fibre –

An optical fiber is cylindrical in shape with three coaxial regions namely
a) Core **b) Cladding** **c) Sheath (Buffer Jacket)**



1. Core:

It is the innermost cylindrical region of an optical fiber also known as light guiding region. Light propagate mainly along the core of the fiber. It has higher refractive index than that of cladding.



2. Cladding:

The core is surrounded by a layer of material called as the cladding. It has less refractive index than that of core. Its function to put the light in the zigzag path by the process of total internal reflection and confine the maximum light to the core.

3. Sheath (Buffer Jacket):

It is the outermost region which acts as a protective layer for optical fiber. It protect the core and cladding from harmful atmospheric effects and physical damage. Moreover, it imparts mechanical strength to the optical fiber.

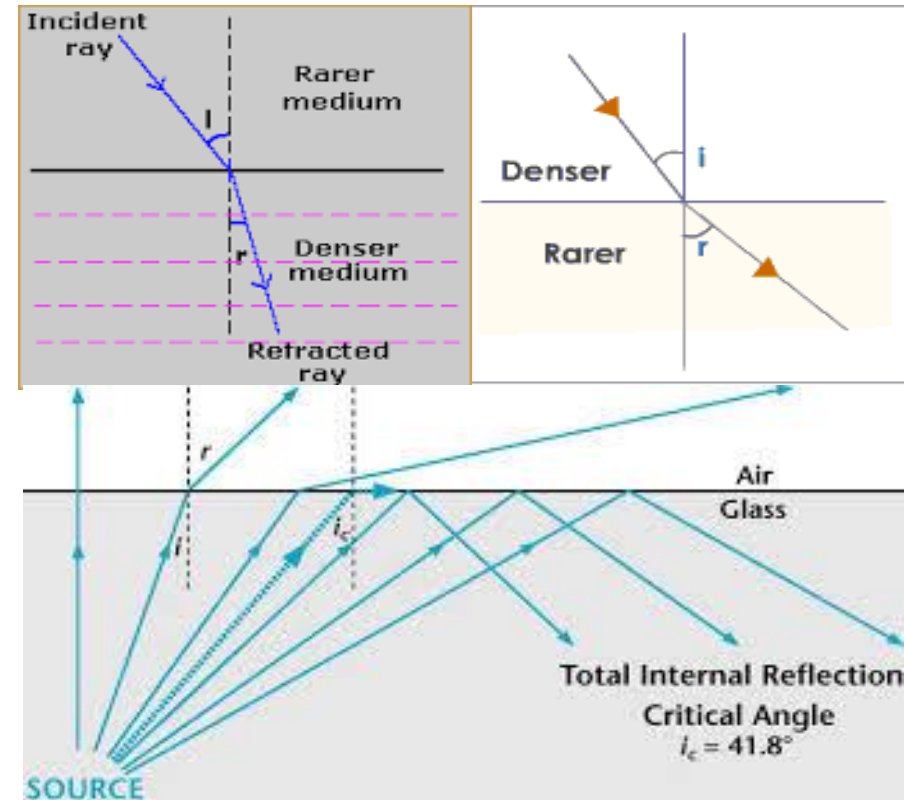
Working Principle of Optical Fiber (Total Internal Reflection)

Whenever ray of light travels from rarer medium to the denser medium, it bends towards to the normal, so angle of refraction r will be less than angle of incident ($\angle r < \angle i$).

But when a ray of light travels from a denser medium to rarer medium, it moves away from the normal. Thus angle of reflection will be always greater than the angle of incident ($\angle r > \angle i$).

If we further increase the angle of incident i , the refracted ray bend more away from the normal such that the angle of refraction is 90° .

The angle of incidence i at which refracted ray makes a 90° angle with the normal is known as critical angle ϕ_c .



The reflection when angle of refraction $r \geq 90^\circ$ is called as Total Internal Reflection. It occurs only when light ray travels from denser to rarer medium.

For $i > \phi_c$, Total Internal Reflection takes place and both, incident ray and reflected ray travel in the same medium.

Expression for Acceptance Angle:

Main function of optical fiber is to collect as much light as possible from the source and transmit it without much loss.

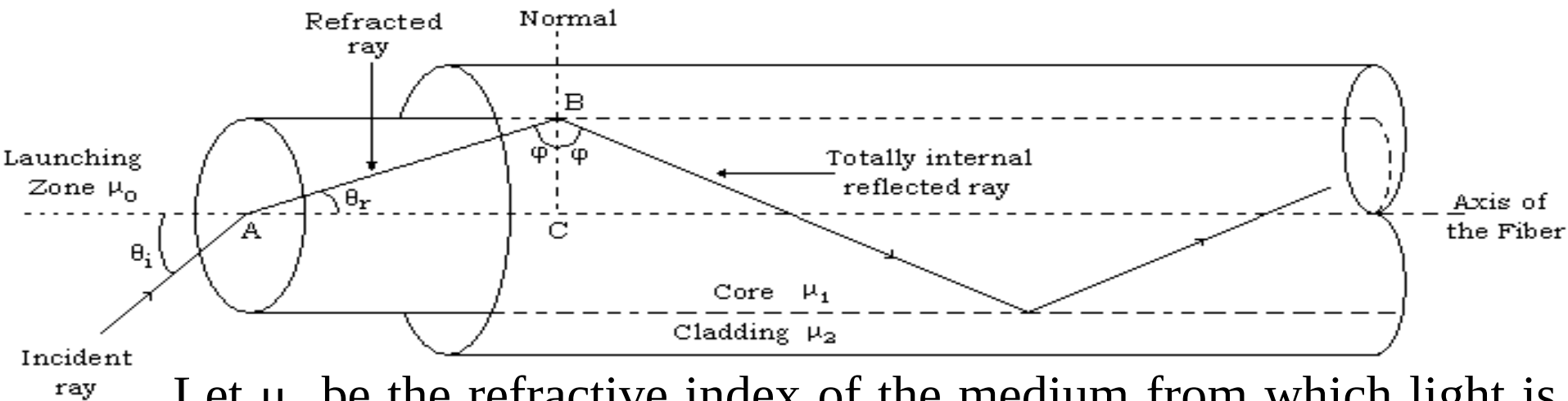
Light collecting capacity of a fiber is depends on two terms.

1. Size of core
2. Numerical Aperture (NA)

Numerical Aperture is depends on

- (a) Acceptance Angle
- (b) Fractional Refractive Index Change

Consider a step index optical fiber in to which light is launched at one end as shown in below fig. The end at which light enters the fiber is called the launching end. Let the refractive index of the core be μ_1 and the refractive index of the cladding be μ_2 such that $\mu_1 > \mu_2$.



Let μ_0 be the refractive index of the medium from which light is launched into the fiber. The ray refracts at an angle θ_r and strikes the core-cladding interface at an angle ϕ . If ϕ is greater than critical angle ϕ_c , the ray undergoes total internal reflection at the interface, since $\mu_1 > \mu_2$. As long as the angle ϕ is greater than ϕ_c , the light ray will stay within the fiber.

Applying Snell's law at the launching end of the fiber, we get

$$\frac{\sin \theta_i}{\sin \theta_r} = \frac{\mu_1}{\mu_0} \dots \dots \dots (1)$$

If θ_i is increased beyond a limit, ϕ will drop below the critical value ϕ_c and the ray escapes from the side walls of the fiber. The largest value of θ_i occurs when $\phi = \phi_c$.

From ΔABC , it is seen that

$$\theta_r + \Phi + 90^\circ = (180^\circ)$$

$$\Rightarrow \theta_r = (90^\circ - \Phi)$$

$$\sin \theta_r = \sin(90^\circ - \Phi) = \cos \Phi \quad \dots \dots \dots (2)$$

Using eqⁿ (2) in eqⁿ (1), we obtain

$$\frac{\sin \theta_i}{\cos \Phi} = \frac{\mu_1}{\mu_0}$$

$$\sin \theta_i = \left(\frac{\mu_1}{\mu_0} \right) \cos \Phi$$

when $\Phi = \Phi_c$ than $\theta_i = \theta_{i(max)}$

$$\sin \theta_{i(max)} = \left(\frac{\mu_1}{\mu_0} \right) \cos \Phi_c \quad \dots \dots \dots (3)$$

when $\Phi = \Phi_c$, for refraction at point B, applying snell's law

$$\frac{\sin \theta_i}{\sin \theta_r} = \frac{\mu_2}{\mu_1}$$

Here, $\theta_i = \Phi_c$ (critical angle) & $\theta_r = 90^\circ$ for TIR

$$\frac{\sin \Phi_c}{\sin 90^\circ} = \frac{\mu_2}{\mu_1}$$

$$\sin \Phi_c = \frac{\mu_2}{\mu_1} \quad \dots \dots \dots (4)$$

$$\text{using } \sin^2 \Phi_c + \cos^2 \Phi_c = 1$$

$$\cos^2 \Phi_c = 1 - \sin^2 \Phi_c$$

From eqⁿ (4)

$$\cos^2 \Phi_c = 1 - \left(\frac{\mu_2}{\mu_1}\right)^2$$

$$\cos^2 \Phi_c = \frac{\mu_1^2 - \mu_2^2}{\mu_1^2}$$

$$\cos \Phi_c = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_1} \dots \dots \dots (5)$$

From eqⁿ (3)

$$\sin \theta_{i(max)} = \left(\frac{\mu_1}{\mu_0}\right) \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_1}$$

$$\sin \theta_{i(max)} = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0} \dots \dots \dots (6)$$

Where μ_0 = Refractive Index of launching medium

μ_1 = Refractive Index of Core

μ_2 = Refractive Index of Cladding

$\theta_{i(max)}$ = Maximum value of angle of incident so that $\Phi \geq \Phi_c$
(Acceptance angle)

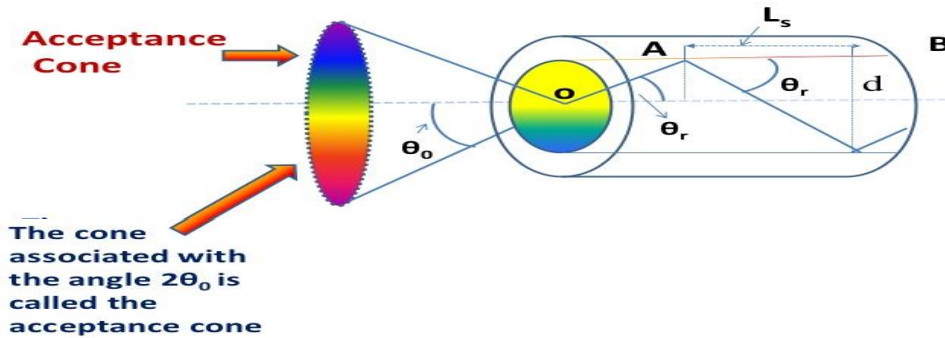
For air $\mu_0 = 1$

$$\sin \theta_{i(max)} = \sqrt{\mu_1^2 - \mu_2^2} \dots \dots \dots (7)$$

$$\theta_{i(max)} = \sin^{-1} \sqrt{\mu_1^2 - \mu_2^2} \dots \dots \dots (8)$$

The angle $\theta_{i(max)}$ is called as acceptance angle of the fiber. It is a maximum angle that a light ray can make with the axis of fiber for its propagation by total internal reflection.

Acceptance Cone



The cone inclosing all the incident directions of light rays for the total internal reflection through the optical fiber is called as acceptance cone ($2\theta_0$).

Fractional Refractive Index Change:

It is the ratio of difference between refractive indices of core and cladding to the refractive index of core. It is denoted by Δ .

$$\Delta = \frac{\mu_1 - \mu_2}{\mu_1}$$

This parameter is always positive because μ_1 must be larger than μ_2 for total internal reflection condition. Typically, Δ is of the order of 0.01.

Numerical Aperture (NA): The numerical aperture is defined as the sine of the acceptance angle. Thus,

$$NA = \sin\theta_0$$

We know that

$$NA = \sqrt{\mu_1^2 - \mu_2^2} \dots \dots (1)$$

$$\mu_1^2 - \mu_2^2 = (\mu_1 - \mu_2) \cdot (\mu_1 + \mu_2)$$

multiply & divide by $2\mu_1$

$$\mu_1^2 - \mu_2^2 = \left(\frac{\mu_1 + \mu_2}{2}\right) \cdot \left(\frac{\mu_1 - \mu_2}{\mu_1}\right) \cdot 2\mu_1$$

Approximating, $\frac{\mu_1 + \mu_2}{2} \approx \mu_1$,

we can express the above relation as

$$\mu_1^2 - \mu_2^2 = 2\mu_1^2 \cdot \Delta$$

$$NA = \mu_1 \sqrt{2\Delta}$$

NA determines the light gathering ability of the fiber. It is a measure of amount of light that can be accepted by a fiber. It is seen from eqⁿ that NA is dependent only on the refractive indices of the core and cladding materials. Its value ranges from 0.13 to 0.50. A large NA implies that a fiber will accept large amount of light from the source.

Mode of Propagation of Light in Optical Fiber:

For the propagation of light by TIR through fiber, it is not true that all such rays which launch by acceptance cone will be propagate along the fiber.

The rays entering the core in direction which is enclosed in an acceptance cone will travel two different paths.

1. Along the axis of fiber
2. Zigzag path

a. Axial Rays:

These are the rays which travel along the axis and hence called as axial rays (at $\phi = \phi_c$).

a. Zigzag Rays:

The rays which follow zigzag path are called as zigzag rays (at $\phi > \phi_c$).

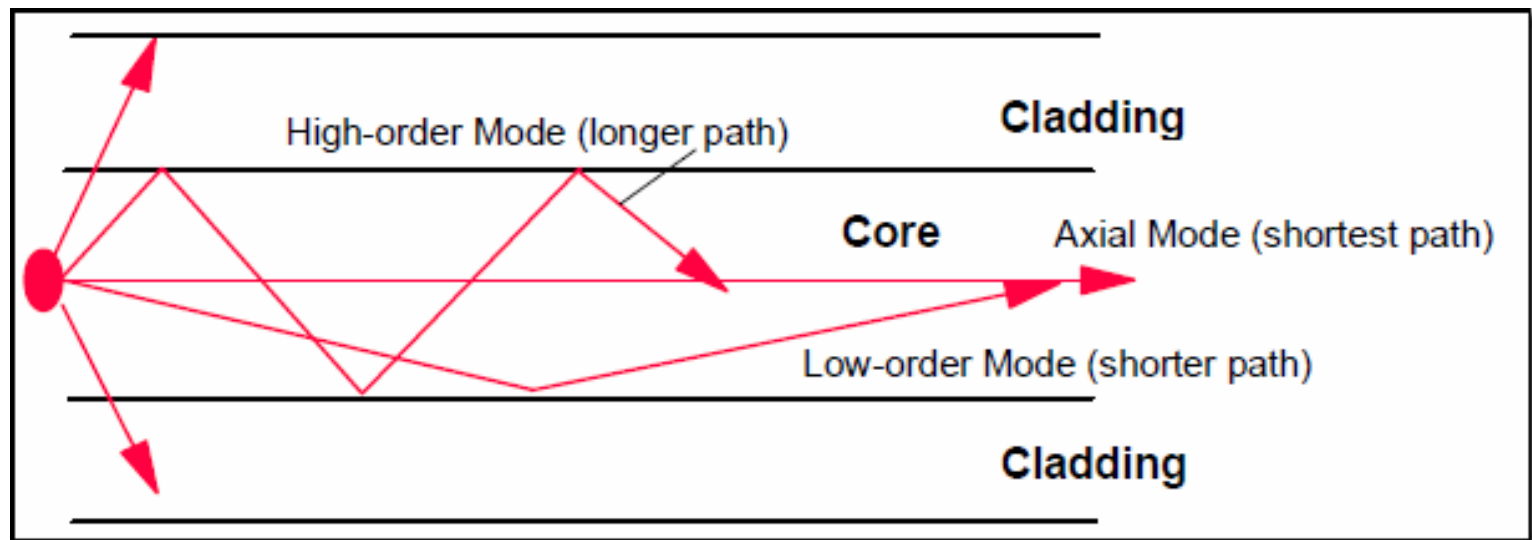
The rays which follow zigzag path repeatedly by the multiple reflection from side walls of optical fiber, phase shift occurs and hence rays travelling along certain zigzag paths may in phase with each others. Such rays interfere constructively and are intensified.

Along certain zigzag paths the rays will be out of phase with each other and hence interfere destructively and get vanish.

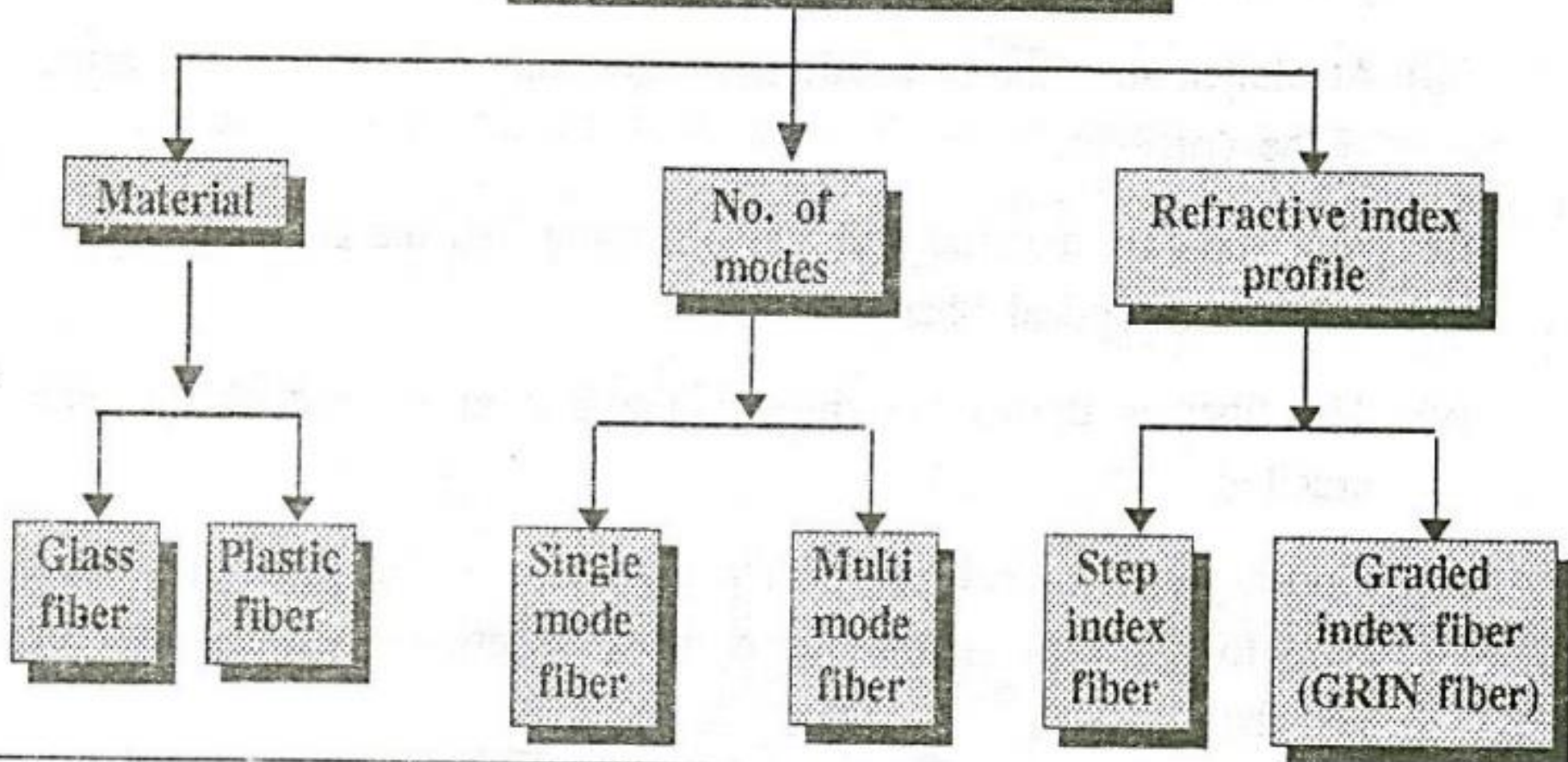
Modes:

The no. of possible paths of light along which ray remain in phase are called as modes of fiber.

No. of possible paths or modes are dependent upon the ratio of d/λ where d = diameter of core, λ = wavelength of light.



Optical fibers are Based on



Index Profile: It is a plot of refractive index drawn on horizontal axis versus the distance from the core on the vertical axis.

It is a graphical representation of the refractive index of the core.

Classification on the basis of Refractive index profile

Optical fibers are classified into two categories on the basis of refractive index profile:

1. Step Index Fiber
2. Graded Index Fiber

Classification on the basis of modes of light propagation

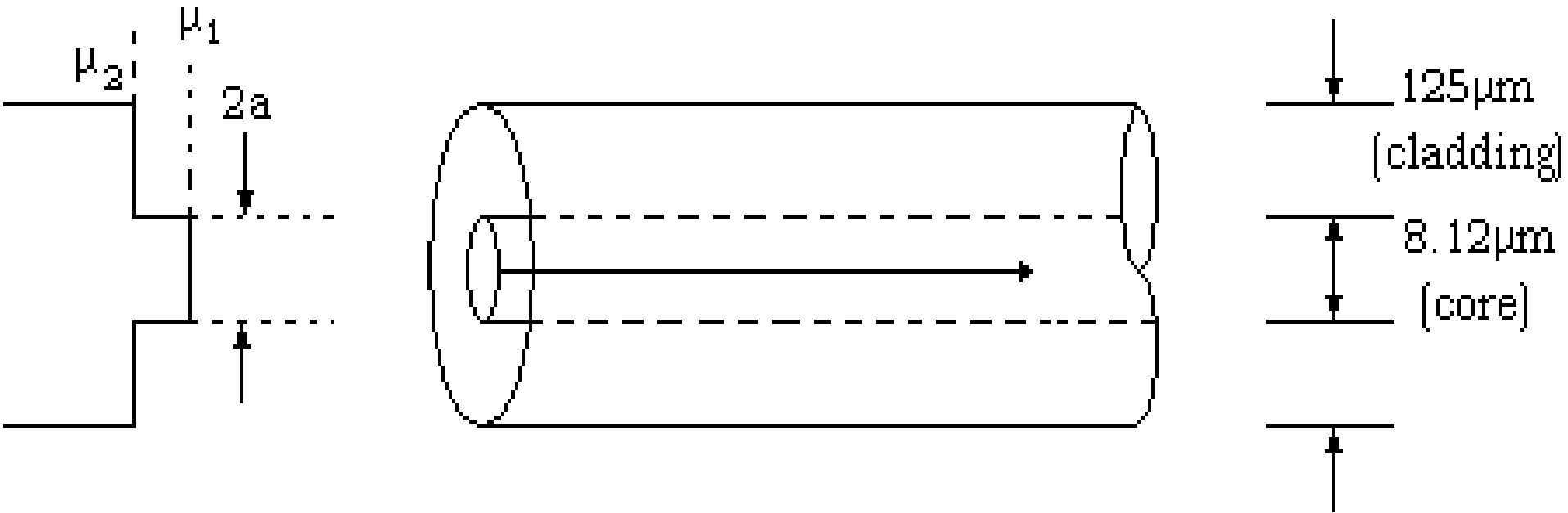
On the basis of modes of light propagation optical fibers are classified into two categories

1. Single Mode Fibers
2. Multi Mode Fibers

Thus optical fibers are classified into three types:

1. Single Mode Step Index Fiber (**SMF**)
2. Multi Mode Step Index Fiber (**MMF**)
3. Graded Index (Multi Mode)Fiber (**GRIN**)

Single Mode Step Index Fiber (SMF):



- It has a smaller core diameter in the range of 8-12 μm . Fiber is surrounded by an opaque protective sheath.
- The Refractive index of the fiber changes abruptly at the core-cladding boundary hence called Step Index Fiber.
- Light travels along a single path i.e. along the axis called zero order mode.
- Δ and NA are very small.

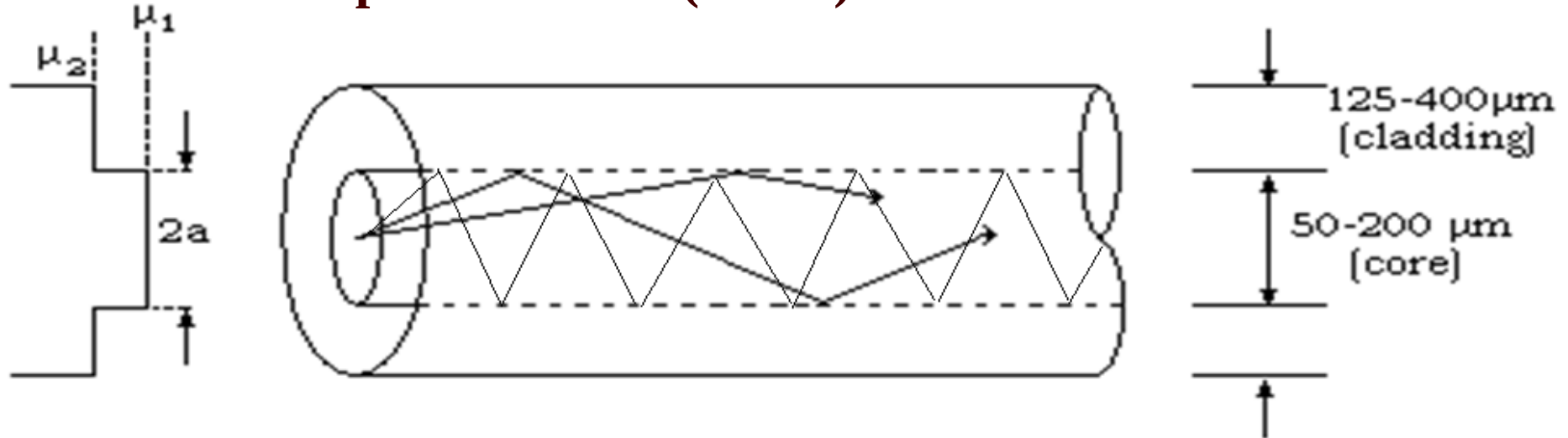
Advantage:

- Intermodal dispersion does not exist in SMF as only one mode exists.
- Low dispersion make the fiber suitable for use with high data transfer rate.
- With careful choice of material, dimension and wavelength, total dispersion can be made extremely small.

Disadvantage:

- Only one mode is possible in SMF.
- NA is small means less amount of light is propagated.
- Acceptance angle is low due to which light coupling in to fiber becomes difficult.
- Manufacturing and handling is more difficult and therefore fiber is costly.

Multimode Step Index Fiber (MMF):



- It is similar to single mode step index fiber but its core diameter is large i.e. of the order of $50-200\ \mu\text{m}$ which is very large compared to the wavelength of light.
- A large no. of modes are possible through MMF.
- Light follows Zigzag paths inside the fibers. Many such zigzag paths of propagation are permitted in MMF.

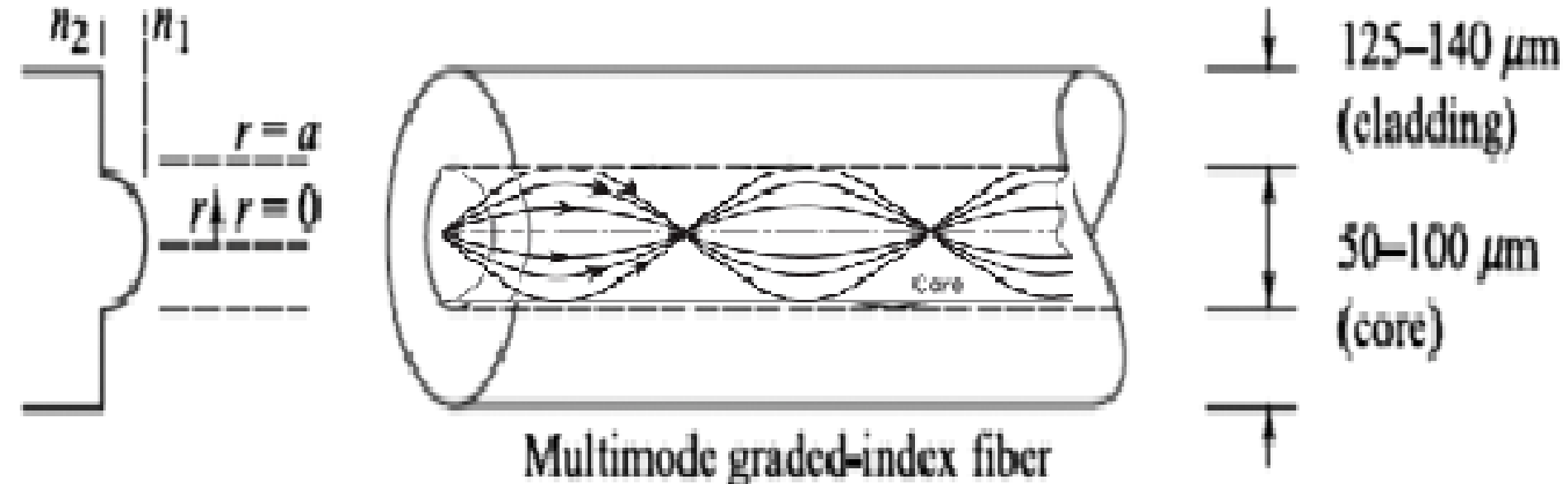
Advantage:

- Many modes can propagate through MMF as core diameter is large.
- It has Large NA of the order of 0.3 thus more light can propagate.
- MMF is relatively easy to manufacture and it is less costly.

Disadvantage:

- Due to propagation of large no. modes through MMF, dispersion is high. Dispersion is Intermodal.
- Higher dispersion means lower data transfer rate and less efficient in transmission.

Graded Index Fiber (GRIN):



- It is a multimode fiber with a core consisting of concentric layers of different refractive indices hence refractive index of the core varies with distance from the fiber axis. It has large value at the Centre and decreases with the increasing radial distance.
- Acceptance angle and Numerical aperture decrease with radial distance from the axis.
- Number of modes is about half than that in a similar multimode step index fiber.
- The size is same as step index fiber.

Advantage:

- GRIN Fiber index profile causes a periodic focusing of light propagating through the fiber.
- The lower number of modes in GRIN Fiber results in lower dispersion than it found in MMF.

Disadvantage:

- The Manufacturing of a GRIN fiber is more complex.
 - Effective acceptance angle of GRIN fiber is less than that of a equivalent SI fiber. It makes coupling fiber to the light source more difficult.
-

Attenuation:

As the light propagate through the fiber , the intensity of light decreases over the length of fiber. This process is known as attenuation.

It is also defined as the ratio of the optical output power to the input power over a fiber length L. it is denoted by α .

If P_i is the optical power launched at the input end of the fiber and P_o is output power of the fiber than attenuation is given by

$$\alpha = \frac{10}{L} \log_{10} \left(\frac{P_i}{P_o} \right) \text{ dB/Km}$$

In an ideal fiber $P_i = P_o$ and $\alpha = 0$.

V-NUMBER:

The number of modes that can propagate through an optical fiber is given by a parameter is called V-number. V-number is generally called as normalized frequency of the fiber. It is given by the relation,

$$V = \frac{2\pi a}{\lambda} (NA)$$

$$V = \frac{2\pi a}{\lambda} \cdot \sqrt{\mu_1^2 - \mu_2^2}$$

$$V = \frac{2\pi a}{\lambda} \cdot \mu_1 \sqrt{2\Delta}$$

Where a is the radius of the core and λ is the wavelength.

- Maximum no. of modes N_m supported by an optical fiber will be dependent upon V-number.
- For a SI fiber is Maximum no. of modes N_m given by,

$$N_m = \frac{1}{2} V^2$$

- For a Graded Index fiber is Maximum no. of modes N_m given by,

$$N_m = \frac{1}{4} V^2$$

- For $V < 2.405$ fiber support only one mode (SMF).
- For $V > 2.405$ fiber support multimode (MMF).

Cut off Wavelength: It is the value of wavelength at $V = 2.405$ for a fiber and it is represented by λ_c

$$\lambda_c = \frac{\lambda V}{2.405}$$

Applications of optical fibers:

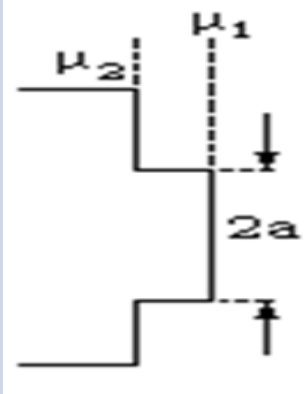
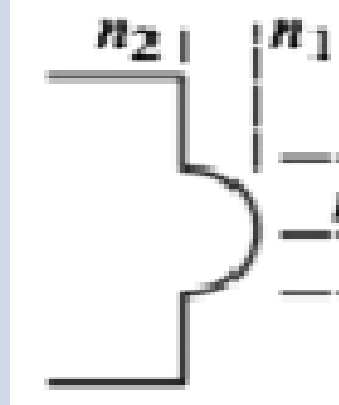
1. Optical fibers are used for illumination and short distance transmission of images.
2. Optical fibers are used as wave-guides in telecommunications.
3. They are used in fabricating a new family of sensors.
4. Fiber optics technology is used in medical diagnostics as well as in medical procedures. The fiber optics endoscope is used to inspect internal organs for diagnostic purposes. In ophthalmology, a laser beam guided by optical fibers is used to reattach detached retinas and to correct defective vision.
5. In cardiology, optical energy transmitted through an optical fiber is used to evaporate built up plaque that is blocking an artery.
6. In treatment of cancer also the optical fiber technology is used.

7. Military applications: An aircraft, a ship or a tank needs tons of copper wires for wiring of the communication equipment; control mechanisms etc. Use of optical fiber in place of copper wire reduces weight and maintains true communication silence to the enemy.
8. Fiber guided missiles are used in recent wars. Sensors mounted on the missile transmit video information through the optical fiber to a ground control van and receive commands from the van again.

Difference between SMF & MMF:

Sr. No.	Single Mode Fiber	Multi Mode Fiber
1	Only one mode can propagate through fiber	More than one mode can be propagated
2	It has smaller core diameter in order to accommodate a single mode.	Higher core diameter to accommodate than one mode.
3	As only one mode travels through it, practically there is no dispersion	Intermodal dispersion occurs in step index fiber but is eliminated using a Graded index fiber.
4	Single mode fibers are more suitable for communication.	Multimode fibers are less suited for communication as there is some sort of dispersion.
5	As the core diameter is very small, joining two fibers is difficult and also launching of light inside the core is very difficult.	As the core diameter is large compared to single mode fiber, joining two fibers or launching of light inside the core is not difficult process.

Difference between Step Index Fiber & Graded Index Fiber:

Sr. No.	SI Fiber	GRIN Fiber
1	Refractive Index of core μ_1 is uniform and that of cladding also uniform.	R. I. of cladding is uniform but that of core is non uniform.
2	As $\mu_1 > \mu_2$, when a ray of light travel from core to cladding, it experience an abrupt change in R. I. at the boundary between core & cladding.	R. I. of core is non uniform, it is maximum at axis of fiber and decreases gradually away from the axis. Thus light ray doesn't experience sudden change in R. I. when it travel from core to cladding.
3	The light ray moves along zigzag path.	The light moves around the axis following a helical path.
4	SI fiber supports single and multimode propagation of light.	GRIN Fiber supports multimode propagation of light.
5		

Advantages of optical fibers in communication:

- 1. Cheaper:** Optical fibers are made from silica which is one of the most abundant materials on the earth. The overall cost of a fiber optic communication system is lower than that of an equivalent cable communication system.
- 2. Smaller in size, lighter in weight, flexible yet strong:** The cross section of an optical fiber is about a few hundred microns. Hence, the fiber cables are less bulky. Optical fibers are quite flexible and strong.
- 3. Not hazardous:** A wire communication link could accidentally short circuit high voltage lines and the sparking occurring thereby could ignite combustible gases in the area leading to a great damage. Such accidents cannot occur with fiber links since fibers are made of insulating materials.
- 4. No Cross Talk:** The light waves propagating along the optical fiber are completely trapped within the fiber and cannot leak out. Hence possibility of cross talk is minimized when optical fiber is used. Therefore transmission is more secure and private.

5. **Wider Band Width:** Optical fibers have ability to carry large amount of information. While a telephone cable composed of 900 pairs of wire can handle 10,000 calls, one millimeter optical fiber can transmit 50,000 calls.
6. **Low loss per unit length:** The transmission loss per unit length of an optical fiber is about 4dB/km.

Optical fiber in communication system:

Optical fiber communication system basically consists of :

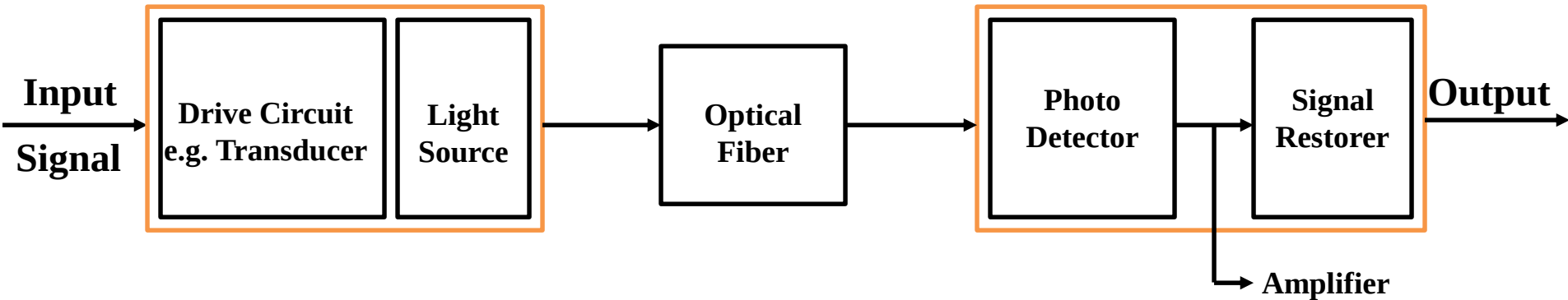
1. Transmitter,
2. Information Chanel,
3. Receiver



The function of :

- 1. Transmitter:** To transfer the input signal to the optical fiber in the form of light.
- 2. Optical Fiber:** To propagate the optical signal without much losses and is passed on to receiver.
- 3. Receiver:** To receive signal from optical fiber in the form of light and to convert it into original form with the help of proper circuit.

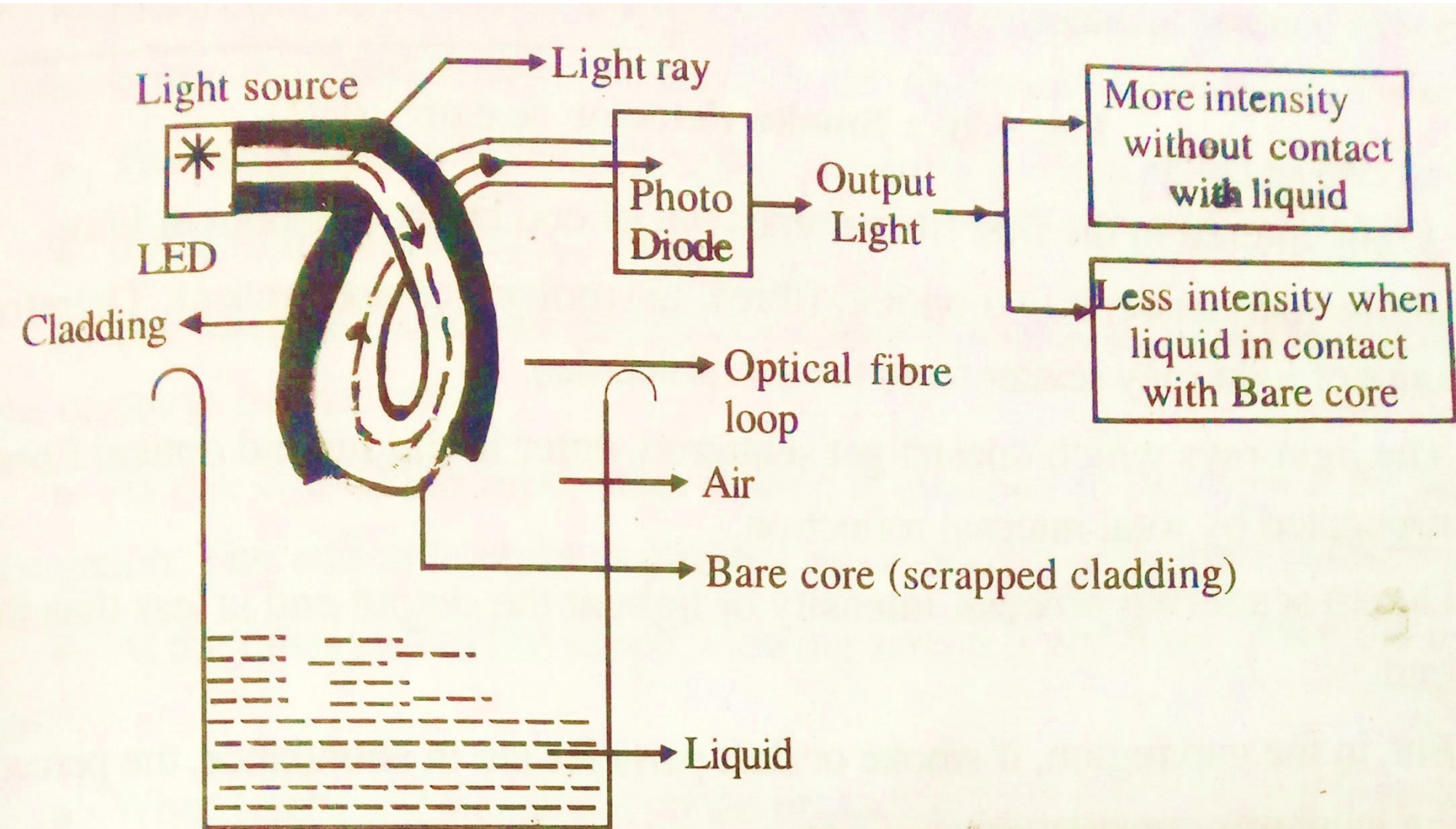
Block diagram of communication system:



- Transmitter consists of a light source with proper drive circuits. Usually the information to be communicated is a non electrical signal. Transducer present in the transmitter converts the non electrical information into an electrical signal.
- The electrical signal thus obtained will have very low frequency and for its propagation it is required to be impressed upon high frequency carrier wave generated by H. F. oscillator. This process of imposing message signal onto a carrier wave is known as modulation. Modulation can be done by either analog or digital technique.
- The signal is transported from transmitter through an optical fiber and is received by receiver.

- The receiver consists of photodetector (semiconductor photodiode), to detect the signal which arrives in the form of light.
- If the signal is weak, it is amplified by an amplifier in the receiver circuit.
- The signal is then fed to the signal restorer which converts light signal in to electrical signal.
- The receiver thus, demodulates the signal for separating the carrier wave and message.
- The message is then fed to the transducer to convert it in initial nature.

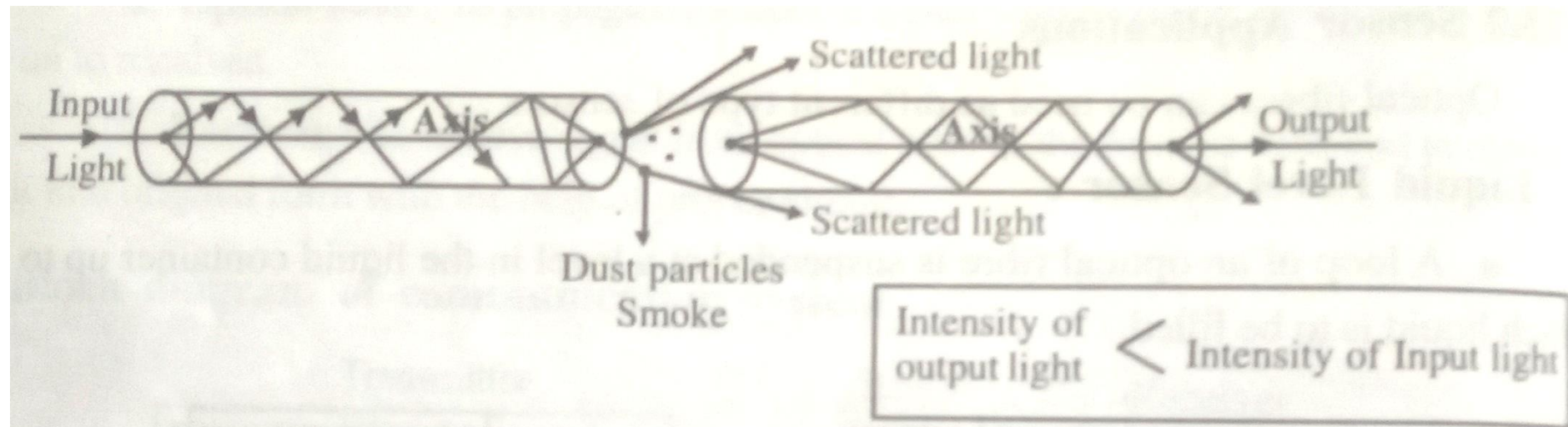
Liquid Level Sensor:



A loop of an optical fiber is suspended at a level in the liquid container up to which liquid is to be filled.

- In this application, small part of cladding is scrapped.
- Light is passed through one end of suspended optical fiber and comes out through another end.
- Usually, LED is used as a light source and a photodetector is used to detect the light at the another end of fiber.
- The intensity of light is measured at output end.
- The intensity of light is more when bare core is in contact with air.
- When level of liquid rises and it touches the optical fiber, the bare core is in contact with the liquid, the intensity of light at output end reduced drastically.
- Sudden decrease in intensity of light at the output end indicates the filling of liquid up to the desired level.
- Such type of sensors are very useful for monitoring the filling of oil tankers, petroleum tank or huge liquid reservoirs.

Smoke Detector or Pollution Detector:

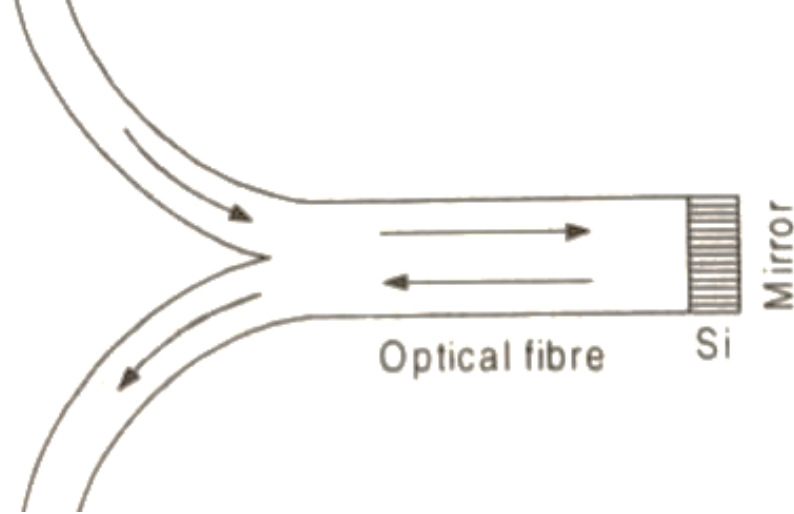


It consists two co-axial optical fibers having a small gap between them as shown in above fig.

- Light entered in the first fiber comes out of end of second optical fiber.
- In the gap between two optical fibers, air molecules are present. Therefore, a small amount of light may scatter from the air molecules.
- The light rays which doesn't get scatter enter in the second optical fiber and get propagated by total internal reflection.
- Due to scattering process, intensity of light at the output end is less than that at the input end.

- But, in the gap region, if smoke or dust particles are in abundance, the percentage of scattering increases considerably.
- The incoming light in the gap region, thus diverges to a great extent and in the second optical fiber. As a result, the intensity of light at the output end decreases drastically when smoke, dust or other pollutants are present in the gap region.
- The reduction in intensity of light at output end is an indication of presence of pollutant in atmosphere.
- The optical fibers thus, can be used as pollution sensing device.

Temperature Sensor:



A thermometer can be built by using:

- (1) Light source – LED
 - (2) Heat sensing element – on optical fiber coil
 - (3) Photodetector – to measure intensity of light.
- When temperature increases, fiber gets heated due to which difference between R. I. of core and that of cladding reduces.
 - The reduction in difference of R. I. of core and cladding will reduce the percentage of light propagating by total internal reflection through optical fiber.
 - Thus more light will leak into the cladding and finally escapes out of fiber.
 - The temperature in the range of 80°C - 70°C can be measured with the help of optical fiber thermometers.

1. Determine the critical angle for a ray traveling from water ($n=1.333$) to air.
2. The core has RI 1.6 and the cladding RI is 1.3. What is the value of critical angle? What is the angle of cone of acceptance ($2\theta_o$)?
3. A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.5 & cladding refractive index of 1.47. Determine a) critical angle at the core-cladding interface b) N. A. for the fiber c) The acceptance angle in the air for the fiber.
4. A typical relative refractive index difference for an optical fiber designed for long distance transmission is 1% Estimate N.A. when the core index is 1.46, calculate the ϕ_c .
7. A single mode step index fiber, which is designed for operation at a wavelength of 1.34mm, has a core and cladding refractive indices of 1.447 and 1.442 resp. when the core diameter is 7.2 mm confirm that the fiber will permit single-mode transmission and estimate the range of wavelength over which this will occur.
8. An optical fiber has a Numerical aperture of 0.2 and a cladding refractive index of 1.59. Determine
 - a) The acceptance angle for the fiber in water which has a refractive index of 1.33,
 - b) The critical angle at the core cladding interface.

9. Calculate the refractive indexes at the core & cladding material of a fiber from the following data $NA = 0.22$, $\Delta = 0.02$.
10. A single mode step index fiber has a core diameter of 7mm and a core refractive index of 1.49. Estimate the shortest wavelength of light, which allows single mode operation when the relative refractive index difference for the fiber is 1%.
11. Compute the maximum radius allowed for a fiber having core refractive index 1.47 and cladding refractive index 1.46. The fiber is to support only one mode at a wavelength of 1300nm.
12. Compute the number of modes supported by a fiber n_1 (core) = 1.54 and n_2 (cladding) = 1.5 core radius 25 mm & operating wavelength is 1300nm.
13. A step index optical fiber of diameter 50 mm has a NA of 0.23. If the wavelength of input light energy is 0.82 mm, determine the number of modes in the cable.
14. Compute the maximum value of $\Delta = (n_1 - n_2) / n_1$ and n_2 R. I. pf Cladding of a single mode fiber having core diameter 10mm and core refractive index 1.5. The fiber is couple to a light source with a wavelength of 1.3 mm. V out off for single mode propagation is 2.405. Also calculate the acceptance angle.

45. A 15km optical fiber link uses fiber with a loss of 1.5 dB/km; the fiber is jointed every km with connectors which give an attenuation of 0.8dB each. Determine the minimum mean optical power which must be launched into the fiber in order to maintain a mean optical power level of 0.3 mw of the detector.
46. A continuous 12 km long optical fiber link has a loss of 1.5 db/km. (1) What is the minimum optical power level that must be launched into the fiber to maintain an optical power level of 0.3 μ W at the receiving end. (2) What is the required input power if the fiber has a loss of 2.5 db/km.